

**Building a Generalizable Deep Reinforcement Learning Model for Trading
Across Diverse Markets**

by

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Dissertation submitted in partial fulfilment of

the requirements for the

Bachelor of Computer Science (Hons)

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Universiti Teknologi PETRONAS

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CERTIFICATION OF APPROVAL

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Approved by,

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UNIVERSITI TEKNOLOGI PETRONAS
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CERTIFICATION OF ORIGINALITY

This is to certify that I am responsible for the work submitted in this project, that the original work is my own except as specified in the references and acknowledgements, and that the original work contained herein have not been undertaken or done by unspecified sources or persons.



WONG XING HAO

ABSTRACT

Current Deep Reinforcement Learning (DRL) models for stock trading have limitation in generalization across diverse financial markets, which it performs well in familiar environments but fail in unseen markets due to market-specific dynamics. This restricts their applicability in real-world, evolving trading scenario. Hence, the objective of this project is to develop a DRL framework that achieves robust generalization, especially in zero-shot or few-shot learning scenarios, by integrating market-agnostic features and advanced techniques like transfer learning and meta-learning to ensure consistent performance, which measured by metrics like the Sharpe ratio, across heterogeneous markets.

The methodology involves collecting historical data (2015–2024) for equities, market indices, and cryptocurrencies (2018–2024) from reliable sources such as Yahoo Finance and Alpha Vantage. Then, market-agnostic features such as price-based metrics, technical indicators, and macroeconomic signals are engineered and normalized. By utilizing the FinRL framework, a DRL environment is developed with these features, diverse trading actions, and a Sharpe ratio-driven reward function. Lastly, multiple DRL algorithms such as Proximal Policy Optimization (PPO), Soft Actor-Critic (SAC), and Deep Q-Networks (DQN) will be experimented with and integrated with transfer and meta-learning techniques to enhance adaptability.

For future work, it involves implementing and training these models, evaluating their generalization across markets, and investigating advanced architectures and additional asset classes like commodities or forex. As concluded, this research holds critical importance for pioneering adaptable trading systems, potentially revolutionizing financial strategies by enabling efficient, scalable performance across diverse markets with minimal retraining.

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CHAPTER 1

INTRODUCTION

1. BACKGROUND

Financial markets are characterized by high volatility, inherent non-stationarity, and dynamic complexities driven by macroeconomic events, geopolitical developments, and varying investor behaviours [1], [2], [3]. Traditional trading approaches, including statistical models like ARIMA and GARCH, as well as classical machine learning methods such as Support Vector Machines (SVM) and Random Forest, generally rely on historical data and static assumptions, limiting their effectiveness during periods of significant regime shifts or unprecedented events [1], [3]. These limitations have intensified the demand for more adaptive and robust strategies that can perform consistently across rapidly evolving market conditions and diverse assets.

In recent years, Deep Reinforcement Learning (DRL), a combination of reinforcement learning and deep neural networks, has emerged as a promising solution to design sophisticated trading strategies [3]. By interacting iteratively with market environments modelled as Markov Decision Processes (MDPs), DRL algorithms can learn sequential actions such as buying, holding, or selling, and aimed at maximizing cumulative rewards with metrics such as risk-adjusted returns like the Sharpe ratio. Empirical evidence has highlighted DRL's superiority in handling complex, non-linear market dynamics when compared to traditional econometric models and static prediction-based methods [1], [3].

Despite having these advancements, a significant challenge for DRL applications in trading is their limited generalization capability across markets or asset classes [1], [4]. Current DRL models frequently exhibit strong performance when trained and tested within similar market conditions (e.g., U.S. equity markets) but suffer performance degradation when exposed to previously unseen market regimes or distinct asset classes such as cryptocurrencies or emerging markets. This shortcoming issue from the non-stationary and heterogenous nature of financial markets, where fundamental dynamics including volatility distributions, liquidity levels, regulatory constraints, trading hours, and investor sentiment can vary significantly [1], [2], [3]. Hence, deploying DRL agents practically often demands intensive retraining and parameter tuning for each new market environment, limiting real-world adaptability and scalability.

Recent research efforts propose overcoming this challenge through innovative learning paradigms, including meta-learning, transfer learning, and cross-domain adaptation. Meta-learning, or "learning to learn," aims to equip DRL models with the ability to rapidly adapt to novel tasks using minimal data which is a process known as few-shot learning or to generalize effectively even without exposure to new environments, capturing zero-shot learning capabilities [3], [4]. Similarly, transfer learning capitalizes on transferring knowledge from one domain to another, allowing models trained on plentiful data in one market to quickly adapt to a different yet related market scenario, significantly reducing demands for retraining and manual tuning [3].

Nevertheless, achieving a truly generalizable DRL model capable of zero-shot or few-shot learning across diverse market conditions remains an open and critical research challenge. Previous frameworks have mostly relied on narrow distributions of tasks or fixed asset universes, neglecting the practical difficulties arising from varying market characteristics, structural divergences, and economic conditions [1], [4]. Additionally, the computational demands and requirement for trading decisions at medium or high frequency further challenge practical implementations.

2. PROBLEM STATEMENT

This study addresses the challenge of developing a DRL model that generalizes across diverse markets for zero-shot or few-shot stock trading. Current DRL approaches struggle to perform effectively in unseen markets such as transitioning from mature stock markets such as U.S. stocks to cryptocurrencies without significant retraining, due to market-specific variations in behaviour and structure [1], [3]. This problem limits their practical usage in dynamic trading environments. The research aims to design a DRL framework with market-agnostic features and explore meta-learning or transfer learning to enable robust performance with minimal or no adaptation, evaluating its effectiveness and efficiency across varied market simulations using metrics like the Sharpe ratio.

3. OBJECTIVE

To develop and evaluate a Deep Reinforcement Learning (DRL) framework for stock trading that achieves effective generalization across diverse financial markets, particularly in zero-shot or few-shot learning scenarios, by utilizing market-agnostic features and advanced learning techniques such as meta-learning or transfer learning, with the aim of maintaining robust performance metrics like the Sharpe ratio when transitioning to new markets such as from U.S. stocks to cryptocurrencies.

4. SCOPE OF STUDY

This study focuses on developing a deep reinforcement learning (DRL) framework for zero-shot and few-shot stock trading, emphasizing robust generalization across diverse markets. The scope covers the implementation and assessment of advanced DRL techniques, combined with meta-learning or transfer-learning methodologies, to ensure adaptability from developed markets to different asset classes and markets.

Historical data for equities and cryptocurrencies will be collected to capture various market conditions. Market-agnostic features will be utilized as inputs to the DRL model to enhance generalization. Besides, the research will evaluate performance using risk-adjusted metrics, including the Sharpe ratio, maximum drawdown, and cumulative returns, against baseline trading strategies. Experiments will assess the transfer of learned policies to new markets, focusing on low-frequency and medium-frequency trading settings due to feasibility constraints.

Lastly, model validation will include back testing across varied market conditions. However, the study excludes high-frequency trading, extensive sentiment analysis, and detailed regulatory considerations due to the limited project timeline and computational resources.

CHAPTER 2

LITERATURE REVIEW

1. EVOLUTION OF ALGORITHMIC AND AUTOMATED TRADING

Financial markets have gone through a significant transformation, transition from manual approaches to automated and algorithmic trading solutions. Computer-driven strategies have improved efficiency, accuracy, and profitability in financial trading [2]. The regulatory measures introduced by the Securities and Exchange Commission (SEC) in 2005, have influenced the growth of high-frequency and algorithmic trading practices [5]. Today, the systematic quantitative trading dominates the securities and futures markets, with systematic procedures making easier programming, back-testing, strategy optimization, and risk management.

However, the existing systems encounter limitations, especially in non-stationary market environments. Traditional algorithmic frameworks are often limited by assumption-based models and static rules, which is not suitable for rapidly evolving market conditions and regime shifts [1], [3].

2. STATISTICAL AND CLASSICAL MACHINE LEARNING

Traditional financial forecasting has largely employed statistical models like ARIMA, GARCH, and VAR models. First, the ARIMA models which is popular but often require additional modifications for practical effectiveness due to non-stationary financial datasets [6], [7]. Next, the GARCH approach which commonly applied for

volatility prediction, still faces challenges regarding volatility clustering and model constraints, hence limiting its adaptability during market shocks or unprecedented conditions [8], [9]. Similarly, the VAR-based models usually focus on linear and stable relationships, causing them suboptimal for capturing complex and dynamic market interactions [10].

Furthermore, classical machine learning methods such as Support Vector Machines (SVM), Random Forest, and XGBoost have demonstrated effectiveness in modelling stock price predictions and assessing financial risks [2]. However, these methods struggle with large-scale datasets, nonlinear parameter optimizations, and adapting dynamically to structural market changes [11], [12].

3. DEEP LEARNING IN FINANCIAL FORECASTING

In recent years, deep learning models such as Convolutional Neural Networks (CNNs), Long Short-Term Memory networks (LSTMs), and Transformer-based model, have emerged as popular approaches due to their capability to capture complex nonlinear patterns in financial time series data [13]. For example, the CNNs can efficiently extract spatial features, while LSTMs perform well with temporal dependencies, and Transformer architectures have demonstrated advanced adaptability due to attention mechanisms [14], [15].

However, the literature has identified significant limitations with deep learning models in trading scenarios. Deep learning networks tend to struggle with non-stationarity and concept drift in situations where statistical characteristics of financial data evolve over time, causing reduce in their predictive accuracy during market regime shifts [1], [3].

4. DEEP REINFORCEMENT LEARNING IN FINANCIAL TRADING

4.1 Fundamentals and Standard DRL Algorithms

In order to address the limitations of traditional predictive approaches, DRL approaches combine sequential decision-making abilities of reinforcement learning with robust, nonlinear feature learning abilities of deep neural networks [1], [3]. The

main objective in a DRL for trading is to maximize the cumulative profits or risk-adjusted returns through optimal sequential decisions, such as buy, sell, or hold [1].

Common DRL algorithms in financial applications include Deep Q-Networks (DQN), Proximal Policy Optimization (PPO), Actor-Critic methods, and Soft Actor-Critic (SAC). Each algorithm has their own unique advantages suited to different trading scenarios and risk management [3], [16], [17]. The effectiveness of these algorithms highlights the potential of DRL models in capturing complex market dynamics and generating robust trading strategies.

4.2 Strengths and Limitations of DRL Approaches

Among the key advantages, DRL models perform well at handling complex nonlinear relationships, sequential decision-making, and adaptability to evolving market circumstances [1], [3]. They show superior performance relative to traditional methods, especially during periods of market volatility or regime shifts [18].

However, DRL models still face critical challenges in generalizability across diverse market environments. The literature highlights their vulnerability to domain shift issues which a scenario where learned strategies significantly degrade in performance when applied to new markets or unseen market conditions [3], [16], [19]. This issue primarily come from DRL models' tendency to overfit specific market conditions encountered during training scenarios, limiting their capability to adapt robustly [3].

5. EMERGING GENERALIZATION TECHNIQUES

5.1 Transfer and Meta-Learning in Trading

In order to improve DRL generalization across financial domains, researchers have investigated various adaptation paradigms such as transfer learning and meta-learning. Where the transfer learning allows models trained on one market (source) to quickly adapt to another (target), and significantly reducing the adaptation requirements [3]. Furthermore, techniques like fine-tuning and parameter sharing have shown promising results in adapting pre-trained DRL agents to new financial markets with minimal retraining [3].

Besides, the meta-learning further enhances fast adaptation capabilities by "learning to learn", optimizing the models to adapt quickly to previously unseen tasks or environments with minimal data. Techniques such as Model-Agnostic Meta-Learning (MAML) provide flexible initialization parameters for rapid adaptation within new financial contexts [4], [20]. Similarly, agile meta-learning models like Reptile and Prototypical Networks lend themselves to effective few-shot prediction tasks in sectors such as cryptocurrency pricing or forex trading [21].

5.2 Few-Shot and Zero-Shot Learning Applications

Specific research gaps remain wide open in zero-shot and few-shot adaptation for financial DRL. While few studies have shown promising potential [21], comprehensive and empirically validated evidence demonstrating robust cross-domain generalization from traditional equities to fundamentally distinct asset classes like cryptocurrencies remains limited [3]. Furthermore, zero-shot approaches where models capable of generalizing to entirely novel scenarios without explicit domain training are particularly under-represented within trading strategies contexts, require for more thorough investigation.

6. MARKET-AGNOSTIC AND MACROECONOMIC FEATURES

The introduction of market-agnostic features such as volatility, liquidity metrics, and correlation-based indicators has increasingly gained significance to enhance cross-domain generalization capabilities in financial DRL frameworks. These universal and domain-independent features can greatly improve DRL model performance consistency across various markets [3], [7].

Moreover, integrating macroeconomic indicators such as GDP, interest rates, inflation rate into DRL frameworks improve understanding of broader economic contexts influencing market behaviours and enhances models' reliability during uncertain macroeconomic environments [1], [3], [7]. At the same time, explicit risk management approaches, such as optimizing for risk-sensitive metrics like Sharpe ratio and Sortino ratio, and controlling maximum drawdowns, have enhanced strategy robustness and operational viability for real-time trading purposes [1], [2], [3], [6], [16].

7. RESEARCH GAPS

Critical gaps persist in current DRL research specifically within the generalizability domain for trading strategies. Existing models mostly focus on environment-specific optimization rather than significant cross-market generalization [3]. Besides, there is limited robust experimentation involving DRL applications transferring effectively without extensive retraining between highly differentiated domains, such as from mature stock markets to emerging or cryptocurrency markets [1], [3], [16].

Moreover, few studies have applied meta-learning methods effectively integrated with DRL models in financial contexts, demonstrating the need to explore these paradigms [3]. Next, the limited research integrating comprehensive sets of market-agnostic and macroeconomic features represents an opportunity for enhancing model generalizability.

Hence, there exists a significant opportunity and need to develop DRL frameworks specifically designed for zero-shot and few-shot adaptation capability across diverse markets. Addressing this gap requires novel architectures, integrated frameworks that leverage both transfer and meta-learning techniques, and careful integration of robust, market-agnostic, and macroeconomic data representation methods [1], [3], [22], [23].

CHAPTER 3

METHODOLOGY

This chapter details out the systematic methodology used to develop and evaluate a generalizable Deep Reinforcement Learning (DRL) framework for trading across diverse financial markets, as mentioned in the research objective. It covers the sequential phases of the study, starts with data acquisition and preparation, moving through feature engineering, DRL environment design, model development incorporating advanced learning techniques, model training, and concluding with a comprehensive evaluation strategy. The specific tools, techniques, and procedures used in each phase are described to ensure clarity and reproducibility.

1. RESEARCH FRAMEWORK

Phase 1: Preliminary Study

Problem Formulation → Literature Review

Output:

Clearly formulated research objectives, identifying the generalization problems of DRL models across diverse markets.

Comprehensive literature review highlighting research gaps particularly related to zero-shot/few-shot learning and market generalization.

Phase 2: Data Collection & Preprocessing

Collect historical financial data:
Equities and market indices (2015–2024).
Cryptocurrency data (2018–2024)

Preprocess data:
Handling missing data through interpolation.
Outlier treatment with winsorization.
Normalization and standardization.

Output:

Integrated and normalized multi-market dataset ready for DRL model training.

Phase 3: Feature Engineering

Price-based features:
Returns, Momentum, Volatility

Volume-based metrics:
Trading volume, Liquidity

Technical indicators:
RSI, MACD, Bollinger Bands

Macroeconomic indicators:
GDP, inflation, interest rates

Correlation-based features

Output:

Engineered feature set aimed at enhancing cross-market generalization.

Phase 4: DRL Environment Design

Set up the DRL environment using FinRL-Meta

Formulate trading scenarios as Markov Decision Process (MDP)

Realistic transaction costs and slippage modelling

Configurable trading frequencies (daily/hourly)

Output:

Functional DRL environment (MDP framework using FinRL) capable of simulating diverse financial trading environments.

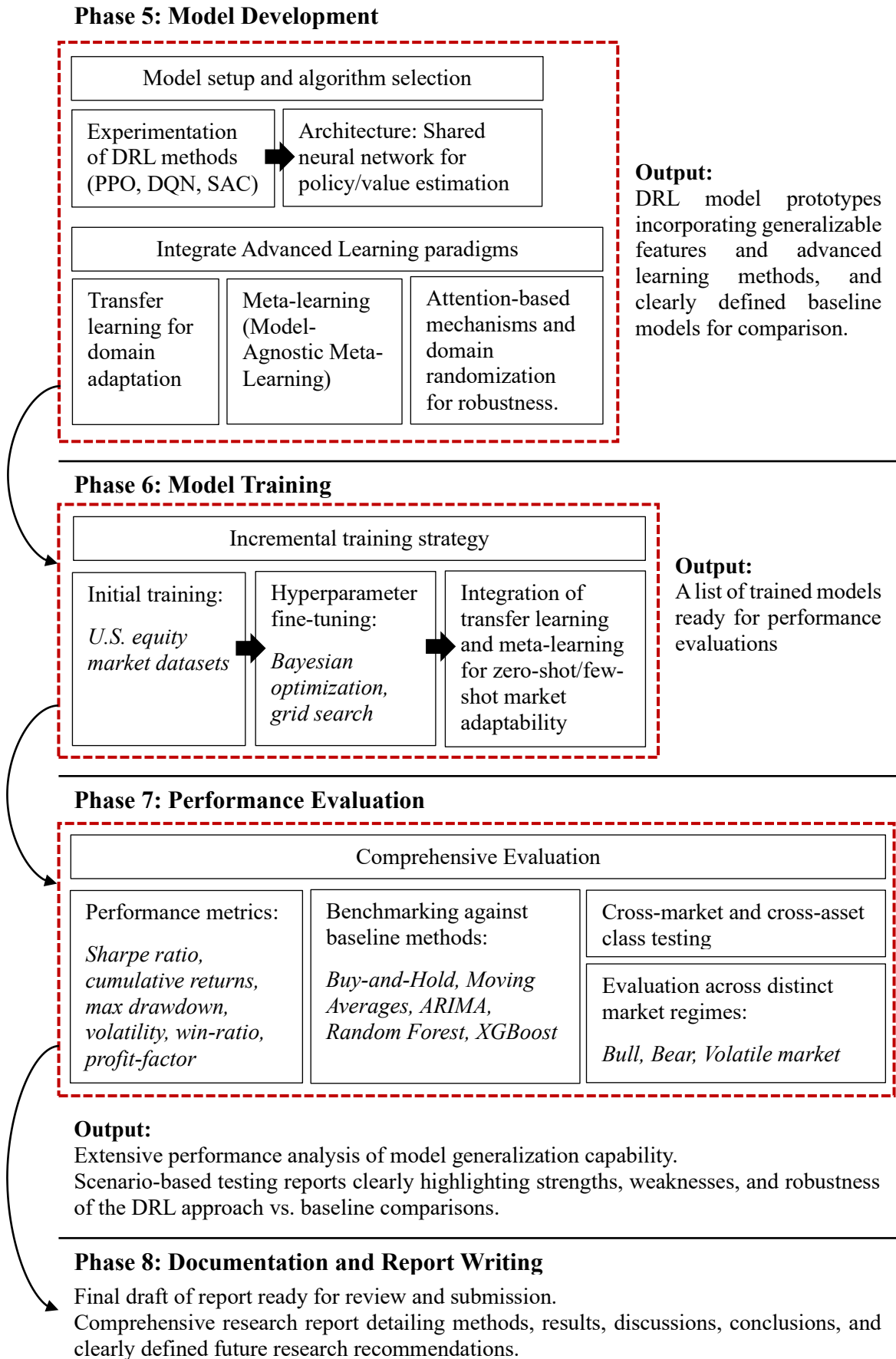


Figure 1 Research Framework

2. DATA COLLECTION AND PREPROCESSING

The very beginning stage is comprehensive data collection and preprocessing, in order to form an accurate, representative dataset covering multiple market conditions. The historical financial data spanning from 2015 to 2024 for equities and market indices, and from 2018 to 2024 for cryptocurrencies, are collected from trustworthy APIs such as Yahoo Finance, Alpha Vantage, Binance, Coinbase, Polygon.io, and Marketstack. These data sources provide critical market data including price (OHLC), volume, indices data and top cryptocurrencies data. While the macroeconomic indicators (interest rates, inflation, and GDP growth) are collected from platforms like Federal Reserve and World Bank, and sentiment indicators are collected from FNSPID [7].

The next step is preprocessing, which going to utilize the FinRL framework that provide functions to efficiently integrating diverse datasets. The preprocessing steps contain synchronization of time series data, interpolation to manage minor gaps, winsorization for outlier treatment, and normalization such as Min-Max scaling and z-score standardization to ensure consistency and enhance model training efficiency.

Table 1 Summary of Data Sources and Time Frame

Data Source	Description	Time Frame
Yahoo Finance, Alpha Vantage	U.S. equities (OHLC, volume, S&P 500)	2015–2024
Binance, Coinbase	Top cryptocurrencies (price, volume)	2018–2024
Polygon.io, Alpha Vantage, Marketstack	Market Indices (emerging markets)	2015–2024
FNSPID	Financial news sentiment	2015–2024
Federal Reserve, World Bank	Macroeconomic Indicators	2015–2024

3. FEATURE ENGINEERING

Market-agnostic feature engineering is one of the important components in generalizing the DRL model across markets. Features such as price-based metrics (returns, volatility, momentum), volume-based metrics (trading volume, bid-ask spreads), technical indicators (RSI, MACD, Bollinger Bands), correlation-based features (asset-market correlation), and macroeconomic signals (interest rates, GDP

growth, inflation) will be collected or generated through FinRL library and standardized using normalization techniques to enable model adaptability across heterogeneous market conditions.

Table 2 Summary of Engineered Features

Feature Category	Examples
Price-Based	Returns, Volatility (20-day rolling std. dev.), Momentum
Volume-Based	Trading Volume, Liquidity, Bid-Ask Spread
Technical Indicators	RSI, MACD, Bollinger Bands
Correlation-Based	Asset-Index Correlations, Market Regimes
Macroeconomic Signals	Interest Rates, GDP, Inflation, Economic surprises

4. DRL ENVIRONMENT DESIGN

For the DRL environment, FinRL-Meta will be the foundation framework to simulate realistic financial markets through a standardized Markov Decision Process (MDP) [16]. The environment will cover diverse asset classes including U.S equities and cryptocurrencies, while integrating transaction costs and slippage to mimic a realistic trading environment. Furthermore, the state space integrates engineered features, while the action space supports discrete (buy, sell, hold) and continuous (proportional portfolio allocation) strategies. Lastly, the risk-adjusted returns such as the Sharpe ratio form the reward function, integrated with maximum drawdown constraints to manage risks effectively. The trading frequency (daily or hourly) is also configurable and adaptable to trading strategies.

Table 3 DRL Environment Components

Component	Description
Framework	FinRL-Meta-based MDP simulation
State Space	Price, technical indicators, macroeconomic signals
Action Space	Discrete (buy, hold, sell), Continuous (scaled positions)
Reward Function	Sharpe ratio; Constrained by maximum drawdown
Trading Frequency	Daily / Hourly
Market Realism	Transaction costs, slippage

5. MODEL DEVELOPMENT

The main DRL framework is developed using Python, PyTorch, and FinRL. The core architecture will be using the Proximal Policy Optimization (PPO) algorithm due to its stability and effectiveness in continuous action environments, and with algorithms such as SAC and DQN variants for comparative analyses. The models adopt an architecture composing a shared neural network structure for policy and value estimation, ensuring robust learning of generalized features across market scenarios.

In order to further enhance market generalization, the developed architecture will incorporate advanced techniques such as transfer learning and meta-learning. For transfer learning, the model will train on U.S stock markets can quickly adapt to cryptocurrencies by partial fine-tuning of deep neural network weights. Furthermore, Model-Agnostic Meta-Learning (MAML) will be used to optimize model initialization, enabling rapid adaptation in few-shot (minimal training data) and zero-shot (no retraining) scenarios, assisting immediate application in diverse environments. Lastly, attention mechanisms and domain randomization might also be applied to further increase model robustness.

6. MODEL TRAINING

Model training involves incremental training starting from U.S. equity market data. Hyperparameters, including learning rate, batch sizes, and discount factors, are tuned utilizing grid search and Bayesian optimization methods, validated by periodic evaluation episodes for early stopping to prevent overfitting.

Then, the post-initial training will introduce market generalization through transfer learning and meta-learning stages. First, in the zero-shot capability tests, the trained model is directly evaluated in unseen cryptocurrency markets. Second, in the few-shot scenarios, the model will undergo fine-tuning on limited cryptocurrency data to evaluate rapid adaptation. Lastly, Meta-learning training (MAML) occurs across multiple market scenarios to establish robust and transfer-efficient model initializations.

7. EVALUATION

Complex assessments use metrics such as the Sharpe ratio, cumulative returns, drawdown, volatility, and trading-level metrics including win-rate, profit factor, and hit ratio. Moreover, benchmark evaluations will include traditional baselines like buy-and-hold and moving-average crossover, and machine-learning baselines (ARIMA, XGBoost, Random Forest). Then, ablation studies are also conducted to identify essential components supporting performance enhancements.

Besides, the cross-market experiments will test the model's adaptability between markets such as equities to cryptocurrencies, across different market regimes such as bull vs. bear environments, and across varying temporal windows to assess consistency and model stability.

Structured evaluations begin with internal validation in U.S. equities followed by comprehensive cross-market testing, ensuring practical applicability and robustness.

Table 4 Evaluation Summary

Aspect	Details
Metrics	Sharpe Ratio, Sortino Ratio, Cumulative Returns, Annual Return, Max Drawdown, Volatility, Hit Ratio, Win-Rate, Profit Factor

Baselines	Buy-and-Hold, Moving Average, ARIMA, XGBoost, Random Forest, Standard DRL, Ablation Studies
Generalization	Cross-Market/Asset Class (Stocks to Crypto), Cross-Regime (Bull to Bear), Temporal (Past to Future markets)

8. PROJECT MILESTONE & TIMELINE

The project duration is roughly seven months with phases clearly split to ensure manageability and feasibility. For the first two months, will cover problem clarification, literature review, and methodology formulation. Then, month 3 is for data acquisition, preprocessing, and development environment setup. Continue with month 4 which involves implementation of feature engineering and baseline models

Reaching month 5, which enter full model development, initial training, and integration of transfer and meta-learning techniques. Then, month 6 will conducts extensive cross-market experiments and performance evaluation. Lastly, month 7 will focuses on final analysis and documentation of results, dissertation compilation, and project completion.

Feasibility risks such as computational demands or overfitting are mitigated by incremental model training, regular validation, tuning, and early stopping procedures. Data sources are also fully publicly accessible, ensuring operational feasibility within the project's allocated timeframe and computational constraints.

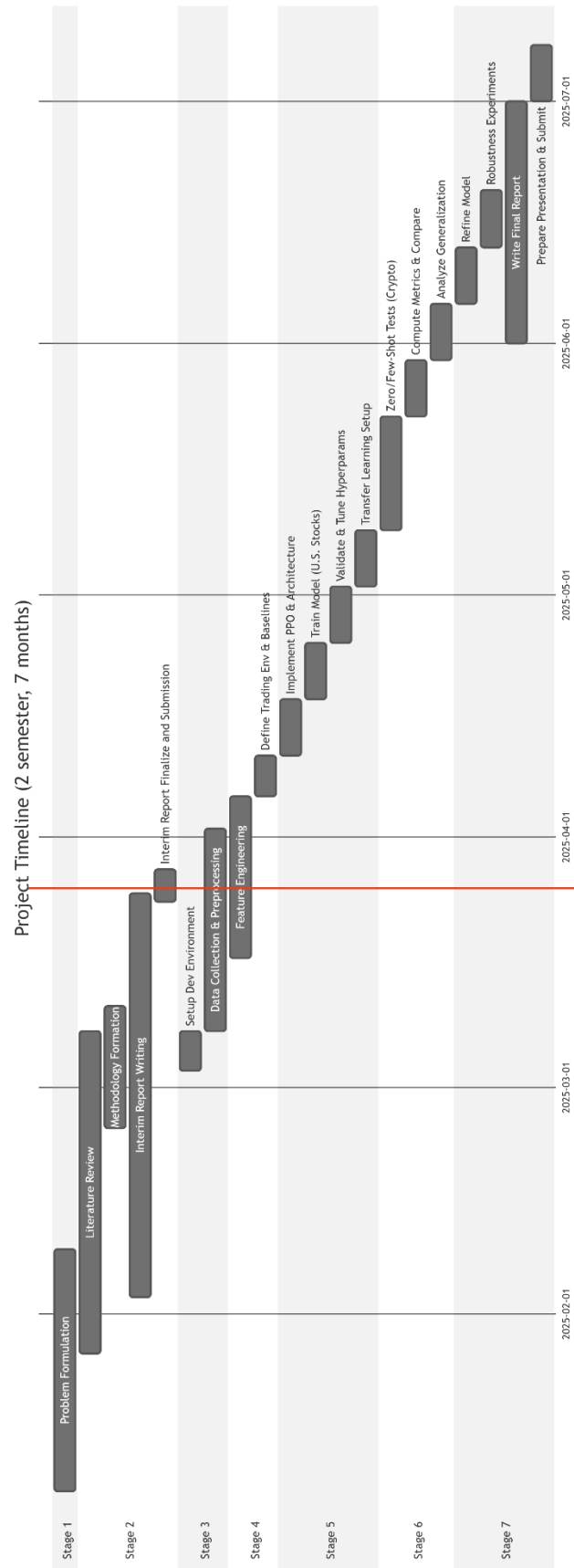


Figure 2 Gantt Chart for Project Timeline

CHAPTER 4

CONCLUSION AND FUTURE WORK

1. CONCLUSION

This project has made some progress in developing a deep reinforcement learning (DRL) framework capable of generalizing across diverse financial markets for stock trading. The initial phases of comprehensive data collection and feature engineering have nearly completed, providing a robust foundation for future development. Historical data spanning 2015 to 2024 for U.S. equities, market indices, and cryptocurrencies (from 2018 onward) were gathered from reliable sources such as Yahoo Finance and Alpha Vantage, ensuring a dataset reflective of varied market conditions. Market-agnostic features including price-based metrics, volume-based metrics, technical indicators, correlation-based features, and macroeconomic signals have been engineered and standardized to support model adaptability.

Furthermore, by leveraging the FinRL framework, the methodology is completed with integration of advanced techniques like transfer learning and meta-learning to facilitate fast adaptation of learned trading policies, enabling effective zero-shot and few-shot trading strategies. After this preparation, the project is ready to proceed with model development and training phases, moving closer to its objective of robust, market-agnostic trading performance.

2. FUTURE WORK

The next steps involve implementing and training a deep reinforcement learning (DRL) model using algorithms like Proximal Policy Optimization (PPO), Soft Actor-Critic (SAC), or Deep Q-Networks (DQN) on a dataset starting with U.S. equity markets. Then, the model's generalization will be tested through zero-shot and few-shot learning in unseen markets, such as cryptocurrencies, with performance evaluated through metrics like Sharpe ratio, cumulative returns, and maximum drawdown and compared against established benchmarks.

Furthermore, future work will include exploring advanced DRL architectures, integrate dynamic risk management, and expand to asset classes like commodities or forex to enhance robustness and scalability, aiming for a powerful trading system adaptable to diverse financial environments with minimal retraining. For the short term, the priority is testing the environment, implementing one algorithm, and evaluating the workflow to ensure it works before jumping into further development.

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